RAJALAKSHMI ENGINEERING COLLEGE

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S2

Started on	Friday, 25 September 2026, 1:27 PM
State	Finished
Completed on	Friday, 25 September 2026, 1:54 PM
Time taken	27 mins 7 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	**g**	**t**	**a**	**b**
s2	**g**	x	**t**	x	**a**	y	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
#include<string.h>
int main(){
    char s1[1000],s2[1000];
    int dp[1001][1001];
    int n,m,i,j;
    scanf("%s",s1);
    scanf("%s",s2);
    n=strlen(s1);
    m=strlen(s2);
    for(i=1;i<=n;i++){
        for(j=1;j<=m;j++){
            if(i==0||j==0){
                dp[i][j]=0;
            }
            else if(s1[i-1]==s2[j-1]){
                dp[i][j]=dp[i-1][j-1]+1;
            }
            else{
                dp[i][j]=(dp[i - 1][j] > dp[i][j - 1])
                        ? dp[i - 1][j]
                        : dp[i][j - 1];

            }
        }
    }
    printf("%d\n",dp[n][m]);
    return 0;

}
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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